

RESPONSE TO REQUEST FOR INFORMATION

RIN 0955-AA13

Accelerating the Adoption and Use of Artificial Intelligence (AI) as Part of Clinical Care

Submitted to:

Department of Health and Human Services (HHS)

Submitted by:

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1.0 Executive Summary

Artificial intelligence holds significant promise for transforming clinical care delivery, improving outcomes, reducing administrative burden, and advancing health equity across the United States. Yet despite the rapid proliferation of AI tools in healthcare settings, adoption remains constrained — not by the absence of capable algorithms, but by the absence of the governance, lifecycle management, and integration infrastructure required to deploy those tools safely, sustainably, and accountably.

This response is submitted by Parish Lemons, President of Total Quality Solutions, LLC, a Missouri-based IT consulting firm with over 20 years of experience managing enterprise software development lifecycles in regulated public-sector environments. The perspectives offered here are grounded in hands-on practitioner experience implementing and governing software systems through the full development lifecycle — including requirements analysis, design, development, testing, deployment, and post-deployment performance management — applied specifically to government and healthcare program contexts.

This submission asserts that the primary barriers to clinical AI adoption are structural and lifecycle-based in nature. Organizations possess increasingly capable AI tools but lack the governance frameworks, evaluation infrastructure, and interoperability standards required to deploy them responsibly at scale. This challenge is especially acute at the state Medicaid program level, where agencies such as Missouri's MO HealthNet division are simultaneously managing legacy system modernization, new federal compliance requirements, and the pressure to integrate AI-enabled solutions — often without the SDLC governance discipline needed to do so safely.

Our recommendations emphasize: (1) establishing SDLC-based governance standards for non-medical device AI; (2) investing in post-deployment monitoring infrastructure; (3) creating reimbursement pathways that incentivize governed AI deployment; and (4) funding cooperative agreements with state Medicaid agencies and mid-market IT consultants to develop and pilot implementation frameworks that the broader health sector can adopt.

2.0 About the Commenter

Total Quality Solutions, LLC is a Missouri-based IT consulting firm providing SDLC leadership, systems modernization, data and reporting solutions, and technology program execution for regulated public-sector environments. The firm supports government and healthcare agencies in translating policy and regulatory requirements into operational, compliant, and auditable technology solutions, with an emphasis on interoperability, reporting integrity, and execution discipline.

Parish Lemons, the firm's President, brings over 20 years of enterprise IT and software development lifecycle experience, including program and project management across complex, multi-stakeholder environments; data integration and analytics enablement supporting operational oversight and program accountability; integration of ERP-based systems within regulated environments; and the development and application of AI-assisted tools to improve data analysis, documentation, and workflow efficiency. Mr. Lemons has supported government procurement activities including RFI and RFP responses, and has pursued active professional development in modernization and data-enabled workflows through leadership and technology cohorts.

Total Quality Solutions brings a practitioner perspective to this RFI that is grounded in the lived reality of implementing technology in regulated environments — a perspective that is essential to informing workable, real-world AI adoption policy.

3.0 Barriers to AI Adoption in Clinical Care

Despite meaningful advances in AI capability, adoption within clinical care environments remains constrained by a set of interrelated, structural barriers. These challenges are not primarily technical in origin — they reflect gaps in governance, lifecycle management, interoperability, and organizational accountability that cut across institutions, vendors, and care settings. From a practitioner SDLC perspective, each barrier maps directly to a phase of the software development and deployment lifecycle that has not been adequately addressed in current AI adoption frameworks.

3.1 Absence of SDLC Governance Frameworks for AI Systems

The most fundamental barrier to responsible AI adoption in clinical care is the absence of established software development lifecycle governance frameworks applied specifically to AI tool deployment and maintenance. In enterprise IT environments, mature SDLC frameworks — including CMMI, ISO/IEC 12207, and Agile-based governance models — provide the structural discipline required to manage software through design, testing, deployment, change management, and retirement. These frameworks define roles, oversight mechanisms,

documentation standards, and escalation procedures that enable organizations to deploy complex software responsibly in regulated environments.

AI tools deployed in clinical settings are software systems. Yet they are routinely introduced outside of SDLC governance structures — evaluated under controlled conditions, deployed without change management protocols, and maintained without version control or performance monitoring plans. The result is a pattern in which clinically promising AI tools are deployed at scale without the institutional discipline required to manage their behavior over time. This gap is not unique to healthcare; it reflects the broader challenge of AI governance in regulated environments, and it is one that HHS is uniquely positioned to address through policy guidance.

3.2 Lack of Post-Deployment Lifecycle Monitoring

Enterprise software development practice has long recognized that the most consequential phase of a system's lifecycle occurs after deployment, when the system encounters the full complexity of real-world operating environments. Post-deployment monitoring, performance trending, drift detection, and structured change management are foundational disciplines in mature software development organizations — and they are largely absent from current clinical AI deployment practice.

AI systems are particularly sensitive to environmental change. Shifts in clinical documentation practices, patient population demographics, care delivery models, or upstream data pipeline configurations can materially alter AI performance over time. Without continuous monitoring and structured performance review cycles, degradation in AI output quality may go undetected until it has already influenced clinical decision-making. This is precisely the category of risk that SDLC governance frameworks are designed to mitigate — and it is a gap that HHS policy can directly address by establishing post-deployment monitoring requirements as a condition of AI tool adoption in federally funded care settings.

3.3 Unclear Accountability Across Multi-Stakeholder Environments

Clinical AI deployments involve a complex web of stakeholders — model developers, data providers, system integrators, health system administrators, clinical end-users, and regulatory bodies — among whom accountability for performance, safety, and remediation is frequently undefined. From a program management perspective, this ambiguity is among the most consequential barriers to responsible adoption. Organizations operating in regulated environments with established risk management obligations are rationally hesitant to deploy AI tools when the governance structure does not clearly define where accountability resides when tools underperform or produce unintended clinical consequences.

This is a governance design problem, not a technology problem. SDLC governance frameworks in enterprise IT environments address this challenge through defined RACI matrices, documented escalation procedures, and contractual performance obligations. HHS can support the extension of these disciplines into clinical AI deployment by issuing guidance that defines accountability expectations for AI tool developers, system integrators, and deploying institutions — particularly in Medicaid and Medicare program contexts where federal funding creates a direct policy lever.

3.4 Workflow Integration and Clinician Trust

AI tools that are not designed for seamless integration into existing clinical workflows impose cognitive and operational burden on clinicians — even when the tools themselves demonstrate strong technical performance under controlled conditions. From an SDLC perspective, this reflects a failure of requirements analysis and user-centered design discipline. Effective enterprise software development begins with deep stakeholder engagement, workflow mapping, and iterative validation with end-users before deployment. These disciplines are inconsistently applied in clinical AI development, resulting in tools that clinicians resist or underutilize.

This challenge is compounded when AI outputs exist outside of the EHR environment, requiring clinicians to leave established workflows to access or interpret results. Integration discipline — ensuring that AI outputs are surfaced within existing clinical workflows and documented within the patient record in a manner consistent with clinical governance requirements — is a solvable engineering and design challenge when treated as a first-class requirement of the SDLC process.

3.5 Data Integrity and Interoperability Gaps

Clinical AI is frequently developed and evaluated using retrospective EHR data that reflects local documentation practices rather than standardized, longitudinal clinical context. Data completeness, consistency, and provenance — disciplines that are foundational to enterprise data governance frameworks — are inconsistently applied in clinical AI development environments. The result is AI tools whose performance is difficult to reproduce across institutions, populations, or care settings.

Interoperability standards such as FHIR have significantly advanced data access and exchange; however, interoperability for AI governance must extend beyond data exchange to encompass metadata, data provenance, evaluation artifacts, and governance policy documentation. Without these additional layers, organizations cannot assess AI performance with the rigor that regulated care environments require. HHS investment in shared data governance infrastructure — including provenance standards, evaluation metadata frameworks, and interoperability requirements specific to AI system inputs and outputs — would materially reduce this barrier.

3.6 State-Level Implementation Constraints: Missouri MO HealthNet

The barriers described above are particularly acute at the state Medicaid program level, where agencies are simultaneously managing the demands of legacy system modernization, new federal compliance requirements, and pressure to integrate AI-enabled solutions — often with constrained resources and limited access to the specialized expertise required to govern AI deployments responsibly.

Missouri's MO HealthNet division exemplifies this challenge. MO HealthNet is currently navigating a multi-year Medicaid Management Information System (MMIS) modernization effort, new federal work requirement verification obligations under the recently enacted Medicaid work requirements framework, and a \$216 million rural health transformation initiative funded by CMS. Each of these programs involves significant IT modernization work — and each creates a potential entry point for AI-enabled tools to improve program administration, eligibility verification, care coordination, and reporting.

Yet Missouri's current technology infrastructure and governance capability are not yet positioned to responsibly deploy AI tools at scale across these programs without the foundational governance, interoperability, and monitoring infrastructure that HHS policy can help establish. The MO HealthNet context is offered here not as an isolated case but as a representative example of the implementation reality facing state Medicaid agencies nationwide — a constituency that HHS policy should explicitly address.

4.0 The Need for Lifecycle-Based Evaluation and Continuous Monitoring

Effective AI adoption in clinical care requires a fundamental shift in evaluation philosophy — from episodic, pre-deployment validation to lifecycle-based assessment that persists through the full operational life of the AI system. This is not a novel concept in software engineering; it is standard practice in mature software development organizations and is directly applicable to clinical AI governance.

4.1 Pre-Deployment Versus Post-Deployment Evaluation

Current practice in clinical AI evaluation is heavily weighted toward pre-deployment testing conducted under controlled conditions. While necessary, pre-deployment evaluation provides limited insight into how AI tools perform when integrated into the full complexity of real-world clinical workflows, diverse patient populations, and variable data quality environments. SDLC governance practice addresses this through structured user acceptance testing, pilot deployment phases with defined success criteria, and post-deployment performance review cycles — disciplines that should be incorporated into clinical AI governance requirements.

4.2 Continuous Performance Monitoring and Drift Detection

AI systems are sensitive to environmental change in ways that traditional software is not. Model drift — the degradation of AI output quality as the operating environment evolves — is a well-documented phenomenon that requires active monitoring to detect and manage. Continuous performance monitoring, including automated drift detection, anomaly flagging, and periodic revalidation cycles, should be established as a standard requirement for AI tools operating in federally funded clinical environments.

HHS can support this requirement by establishing minimum monitoring standards as a condition of Medicaid and Medicare program participation for AI tool vendors, and by investing in shared monitoring infrastructure that state agencies and smaller health systems can leverage without building duplicative internal capability.

4.3 Interoperability Beyond Data Exchange

Interoperability requirements for clinical AI must extend beyond the exchange of clinical data to encompass the full information environment in which AI systems operate. This includes metadata standards for AI input and output documentation, provenance requirements for training and validation data, evaluation artifact standards that enable cross-institutional performance comparison, and governance policy interoperability that allows oversight mechanisms to function across organizational boundaries.

Without these additional interoperability layers, clinical AI deployments will remain fragmented, difficult to evaluate at scale, and resistant to the kind of comparative effectiveness assessment that informed policy and reimbursement decisions require.

5.0 Regulatory and Policy Considerations

The regulatory landscape for clinical AI is characterized by a mismatch between existing frameworks — designed for discrete medical devices and static health IT systems — and the operational reality of non-medical device AI that evolves continuously, operates across organizational boundaries, and influences care indirectly through workflows and information presentation. Addressing this mismatch does not require comprehensive new regulation; it requires targeted policy guidance that brings AI governance into alignment with established software engineering disciplines.

5.1 Adopt SDLC Maturity Frameworks as AI Governance Standards

HHS should consider formally adopting SDLC maturity frameworks — specifically the Capability Maturity Model Integration (CMMI) or equivalent standards — as the governance baseline for

non-medical device AI operating in federally funded clinical environments. These frameworks are well-established, widely understood by the IT vendor community, and directly applicable to the lifecycle management challenges that currently constrain responsible AI adoption. Requiring AI tool vendors to demonstrate SDLC maturity at a defined level as a condition of program participation would provide a practical, verifiable governance standard that does not require development of novel regulatory infrastructure.

5.2 Clarify Accountability and Governance Responsibilities

HHS should issue guidance that explicitly defines accountability expectations for AI tool developers, system integrators, and deploying health institutions — including documentation requirements, escalation procedures, and performance remediation obligations. This guidance should be structured to apply to the full stakeholder ecosystem of a clinical AI deployment, not only to the AI tool developer, and should be designed to integrate with existing compliance frameworks including HIPAA, the CMS Conditions of Participation, and state Medicaid program requirements.

5.3 Establish AI-Specific Requirements Within Medicaid IT Standards

CMS should establish AI-specific requirements within the existing Medicaid IT standards framework — specifically the Medicaid Information Technology Architecture (MITA) framework and the standards and conditions governing MMIS design, development, and implementation. Integrating AI governance requirements into MITA would provide a direct mechanism for extending SDLC-based governance standards to the state Medicaid program context, ensuring that AI tools deployed within Medicaid infrastructure meet minimum governance, monitoring, and accountability requirements as a condition of the enhanced federal matching funds available for Medicaid IT investments.

6.0 Reimbursement Pathways for AI-Assisted Care

Reimbursement policy is among the most powerful instruments available to HHS for shaping AI adoption behavior in clinical care environments. Current reimbursement frameworks create limited incentive for health systems to invest in the governance infrastructure required to deploy AI tools responsibly — because governance investment is not reimbursable, and because reimbursement for AI-assisted care is inconsistent and poorly defined.

6.1 Create Pilot Reimbursement Pathways for Governed AI Deployments

CMS should establish pilot reimbursement pathways — through the Physician Fee Schedule, Medicaid managed care authorities, or CMS Innovation Center models — specifically for AI-assisted care coordination and clinical decision support tools that meet defined governance

standards. Tying reimbursement eligibility to governance requirements creates a direct financial incentive for health systems to invest in SDLC-based AI governance — and provides CMS with a practical mechanism for evaluating the effectiveness of those governance requirements in real-world deployment contexts.

6.2 Align Medicaid Enhanced Match Incentives with AI Governance Standards

The 90% enhanced federal matching rate available for Medicaid IT system design, development, and implementation should be explicitly extended to cover the SDLC governance infrastructure required for responsible AI deployment — including monitoring systems, evaluation frameworks, and interoperability capabilities. States are currently incentivized to invest in IT infrastructure generally, but lack specific guidance on how AI governance investments qualify for enhanced matching. Clarifying this eligibility would materially reduce the financial barrier to responsible AI adoption at the state Medicaid program level — a direct benefit to programs such as Missouri's MO HealthNet that are simultaneously managing modernization and AI integration demands.

7.0 Role of R&D and Public-Private Partnerships

Research and development investment by HHS should be directed toward the foundational infrastructure gaps that constrain responsible AI adoption — not toward additional AI model development, which the private sector is pursuing aggressively and at scale. The comparative advantage of HHS R&D investment lies in creating shared public goods: evaluation frameworks, governance standards, interoperability infrastructure, and implementation playbooks that private-sector entities cannot produce individually.

7.1 Fund SDLC Governance Playbooks for Clinical AI

HHS should fund the development of implementation playbooks that translate SDLC governance disciplines into practical, actionable guidance for clinical AI deployment — specifically designed for use by state Medicaid agencies, community health systems, and mid-market health IT organizations that lack the internal resources to develop governance frameworks independently. These playbooks should address the full AI deployment lifecycle from requirements definition through post-deployment monitoring, and should be developed through cooperative agreements that include practitioner-level IT consultants with direct experience in regulated government and healthcare environments.

7.2 Establish State Medicaid AI Governance Pilots

HHS should fund cooperative agreements with a cohort of state Medicaid agencies to pilot SDLC-based AI governance frameworks within active program modernization contexts. Missouri's MO HealthNet is a strong candidate for such a pilot, given the convergence of active MMIS

modernization, new work requirement verification obligations, and CMS-funded rural health transformation investment that collectively create a rich testing environment for AI governance frameworks. Pilot findings should be published as shared learning resources available to all state Medicaid programs.

7.3 Prioritize Mid-Market and Practitioner Voices in R&D Investment

HHS R&D investment in clinical AI has historically been concentrated in large academic medical centers and major defense and technology contractors. The implementation challenges that constrain AI adoption in community health systems, state Medicaid agencies, and smaller health organizations are distinct from those facing large institutions — and the solutions developed in large institutional contexts are frequently not transferable. HHS should explicitly prioritize mid-market IT consultants, regional health systems, and state agency partners in cooperative agreements and applied research investments, ensuring that AI governance solutions are developed and validated in the contexts where adoption challenges are greatest.

8.0 Closing Recommendations

Total Quality Solutions, LLC respectfully offers the following summary recommendations for HHS consideration in developing policy and programmatic responses to this RFI. These recommendations are grounded in practitioner experience in regulated IT and public-sector environments, and are designed to be actionable, verifiable, and scalable across the diversity of clinical care contexts in which AI adoption is occurring.

Section	Recommendation	Primary Policy Area
Section 3.1	Adopt CMMI or equivalent SDLC maturity frameworks as governance standards for non-medical device AI in federally funded clinical environments	Regulation
Section 3.2	Establish post-deployment monitoring requirements — including drift detection and performance review cycles — as conditions of AI tool participation in Medicaid and Medicare programs	Regulation
Section 3.3	Issue guidance defining accountability, documentation, and escalation requirements across the full stakeholder ecosystem of clinical AI deployments	Regulation
Section 3.6	Explicitly address state Medicaid program implementation constraints in HHS AI governance guidance, including MO HealthNet as a representative case	Regulation / R&D

Section	Recommendation	Primary Policy Area
Section 5.3	Integrate AI governance requirements into the MITA framework and Medicaid IT standards and conditions	Regulation
Section 6.1	Create pilot reimbursement pathways for AI-assisted care tools that meet defined SDLC governance standards	Reimbursement
Section 6.2	Clarify 90% enhanced federal match eligibility for SDLC governance infrastructure investments supporting AI deployment in state Medicaid programs	Reimbursement
Section 7.1	Fund development of SDLC governance implementation playbooks for clinical AI, designed for use by state Medicaid agencies and mid-market health IT organizations	R&D
Section 7.2	Establish state Medicaid AI governance pilots, including Missouri MO HealthNet, to develop and validate governance frameworks in active modernization contexts	R&D
Section 7.3	Prioritize mid-market IT consultants and state agency partners in HHS AI cooperative agreements and applied research investments	R&D

Total Quality Solutions, LLC welcomes the opportunity to provide additional input, engage in follow-up consultation, or participate in any technical working groups HHS convenes in response to this RFI. We are available at the contact information provided on the cover page of this submission.

Appendix A: Question Crosswalk

The following table maps the specific questions posed in HHS RFI RIN 0955-AA13 to the corresponding sections of this response.

RFI Question	Question Summary	Response Section	Policy Area
Q1	What regulatory, payment policy, or programmatic changes should HHS prioritize to incentivize effective AI use in clinical care?	Sections 5.0, 6.0	Regulation / Reimbursement
Q2	How can payment structures be simplified and aligned to support efficient AI technologies?	Section 6.1, 6.2	Reimbursement

RFI Question	Question Summary	Response Section	Policy Area
Q3	What are the biggest barriers to private sector AI innovation in healthcare?	Section 3.0 (all subsections)	Regulation
Q4	What best practices should HHS adopt to support private sector AI activities?	Sections 4.0, 5.1, 5.2	Regulation
Q5	How can HHS invest in R&D to integrate AI in care delivery?	Section 7.0 (all subsections)	R&D
Q6	How can HHS support public-private partnerships for clinical AI?	Sections 7.2, 7.3	R&D
Q7	What interoperability standards are needed for AI in clinical workflows?	Sections 3.5, 4.3	Regulation
Q8	How should HHS address accountability and governance for non-medical device AI?	Sections 3.3, 5.2	Regulation
Q9	How can HHS address equity in clinical AI adoption?	Sections 3.5, 3.6, 7.3	Regulation / R&D
Q10	What role should state Medicaid programs play in AI adoption?	Sections 3.6, 5.3, 6.2, 7.2	All three areas

End of Response — Total Quality Solutions, LLC | solutions@tqsolve.com | 314-605-6519